

Technical Note

Probing the limit of hydrologic predictability with the Transformer network

Jiangtao Liu^a, Yuchen Bian^b, Kathryn Lawson^a, Chaopeng Shen^{a,*}^a Civil and Environmental Engineering, The Pennsylvania State University, United States^b Amazon Search Science and AI, United States

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ABSTRACT

For a number of years since their introduction to hydrology, recurrent neural networks like long short-term memory (LSTM) networks have proven remarkably difficult to surpass in terms of daily hydrograph metrics on community-shared benchmarks. Outside of hydrology, Transformers have now become the model of choice for sequential prediction tasks, making it a curious architecture to investigate for application to hydrology. Here, we first show that a vanilla (basic) Transformer architecture is not competitive against LSTM on the widely benchmarked CAMELS streamflow dataset, and lagged especially prominently for the high-flow metrics, perhaps due to the lack of memory mechanisms. However, a recurrence-free variant of the Transformer model obtained mixed comparisons with LSTM, producing very slightly higher Kling-Gupta efficiency coefficients (KGE), along with other metrics. The lack of advantages for the vanilla Transformer network is linked to the nature of hydrologic processes. Additionally, similar to LSTM, the Transformer can also merge multiple meteorological forcing datasets to improve model performance. Therefore, the modified Transformer represents a rare competitive architecture to LSTM in rigorous benchmarks. Valuable lessons were learned: (1) the basic Transformer architecture is not suitable for hydrologic modeling; (2) the recurrence-free modification is beneficial, so future work should continue to test such modifications; and (3) the performance of state-of-the-art models may be close to the prediction limits of the dataset. As a non-recurrent model, the Transformer may bear scale advantages for learning from bigger datasets and storing knowledge. This work lays the groundwork for future explorations into pretraining models, serving as a foundational benchmark that underscores the potential benefits in hydrology.

1. Introduction

Rainfall-runoff modeling is essential for flood prediction, water resource management, and environmental protection (Hrachowitz and Clark, 2017). Rainfall-runoff modeling is a critical aspect of hydrology, as it models the intricate relationships between precipitation, watershed characteristics, and streamflow. The introduction of long short-term memory (LSTM) networks marked a significant advancement in this field for numerous variables of interest including soil moisture (Fang et al., 2017; Liu et al., 2022b, 2023), streamflow (Botterill and McMillan, 2023; Feng et al., 2020, 2021; Konapala et al., 2020; Kratzert et al., 2019; Sun et al., 2021; Xiang and Demir, 2020), water temperature (Rahmani et al., 2021b; Rahmani et al., 2021a), and groundwater levels (Afzaal et al., 2020; Wunsch et al., 2022). For these applications, LSTM consistently outperformed traditional models and process-based models (Feng et al., 2020; Papacharalampous et al., 2018). LSTM's ability to learn many-step dependencies and handle variable-length input

sequences has proven particularly advantageous in capturing the inherent complexity of hydrological processes (Hochreiter and Schmidhuber, 1997).

As a recurrent neural network (RNN), LSTM processes data sequentially through time steps, updating its internal states at each step based on the current input and the previous states. This iterative process, which involves repeatedly applying its internal neural network mechanisms, leads to some limitations. The recurrent nature means RNNs are prone to an issue called the vanishing gradient (Hochreiter, 1991; Hochreiter et al., 2001), where the gradient of the loss with respect to the network weights becomes very small, making network training extremely slow. This issue limits the length of the training sequence, and reduces the impact of inputs from the longer-term past on present predictions. This could be one of the reasons why baseflow was previously identified as a limitation (Feng et al., 2020). Even though LSTM was developed to mitigate this issue and can suppress it better than the original RNNs, it is not immune to it (Dai et al., 2019; Zhang et al.,

* Corresponding author.

E-mail address: cshen@engr.psu.edu (C. Shen).<https://doi.org/10.1016/j.jhydrol.2024.131389>

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2016). Furthermore, recurrence means these time steps must be taken in sequence — the time steps cannot be run in parallel. This imposes a restriction on the efficiency of parallel processing, and thus the scale of data on which the model can be trained.

In many applications outside hydrology, the Transformer architecture (Vaswani et al., 2017) has demonstrated superior performance over LSTM networks in various domains, including machine translation, speech recognition (Karita et al., 2019), natural language processing and sentiment analysis (Devlin et al., 2019), question answering (Rajpurkar et al., 2018), computer vision (Carion et al., 2020), protein structure prediction (Rives et al., 2021), and music generation (Huang et al., 2018). The Transformer model uses an attention mechanism, where each word (or “input token”) is transformed into three different kinds of information: a ‘query’ that asks how relevant other words are to it, a ‘key’ that responds to others’ queries about its relevance, and a ‘value’ that carries the word’s actual meaning. The model calculates the relevancies between the query and keys of all words, then combines the values of the most relevant words to understand the current word better. With LSTM, the most recent input tokens are always more important than further-away ones, whereas a Transformer could learn to put more focus on further-away tokens (Dehghani et al., 2019; Raganato and Tiedemann, 2018), which makes it ideal for language modeling. Moreover, as it does not have recurrence, a Transformer can run the time steps in parallel and can scale up in parallel computation when more data and more GPUs are available. Considering such benefits, there should be a heightened interest in harnessing Transformers for hydrologic applications. Transformers are increasingly being used in hydrologic and water quality modeling (Castangia et al., 2023; Xu et al., 2021; Yang et al., 2023a; Koya and Roy, 2023; Li and Yang, 2019), especially for near-term forecasting. However, the scale of application tends to be limited and their benchmarking on standardized, well-understood datasets, such as the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) dataset (Addor et al., 2017; Newman et al., 2014), remains limited in the literature. It is thus intriguing whether the Transformer’s advantages over recurrent networks will apply in the case of natural systems, which can be argued to lack the irregular sequential structure found in human languages.

While some past hydrologic studies have claimed superior performance for other architectures compared to LSTM, many times a rigorous comparison was not carried out due to the different modeling objectives. The conclusions were often conditional on using a small dataset for benchmarking (e.g., see Abed et al., 2022; Amanambu et al., 2022; Ghobadi and Kang, 2022), and using procedures and configurations (training and test periods, sites, and forcing data) that differ from published benchmarks (Yin et al., 2022, 2023), or on a case study which was not easy to compare to the work of other independent teams (Liu et al., 2022a; Koya and Roy, 2023). Specifically, Yin et al. (2022) proposed the RR-Former model (a Transformer variant) and conducted experiments with 7-day forecasts on the CAMELS dataset. They modeled 673 distinct basins independently and calculated performance metrics for each, and they also assessed a selected set of 448 basins using a single model. In contrast, our research primarily focuses on long-term prediction problems rather than forecasting. Forecasting typically involves predicting results within a relatively short period based on historical data, whereas our study concentrates on the long-term rainfall-runoff relationship to better understand its underlying patterns. Building on the work by Yin et al. (2022), Yin et al. (2023) introduced the RRS-Former model, which conducted a one-day-ahead runoff experiment. A similar study by Feng et al. (2020) applied a data integration approach to an LSTM model. Although direct comparison is challenging, Feng et al. (2020) reported a median NSE of 0.86, which was superior to the RRS-Former model’s performance in Yin et al. (2023). Koya and Roy, (2023) evaluated the Temporal Fusion Transformer (TFT) model on the Caravan dataset (Kratzert et al., 2023) and reported median Kling-Gupta efficiency (KGE) of 0.705. However, Feng et al. (2023) benchmarked LSTM on a similarly large dataset and showed median KGEs of 0.74 for 3753 global

basins and 0.78 for 1675 basins with long-term records. Furthermore, while more benchmarking is welcomed, the model in Koya and Roy, (2023) is not purely attention-based, as it incorporates some LSTM layers in its encoder, making it difficult to determine whether the performance improvements are due to the attention mechanism or the LSTM layers. This approach also brought back time recurrence and did not leverage the time parallelism of the Transformer network as advocated in the original Transformer model. In the interest of reproducibility and comparability, which underpin scientific progress, it is useful to benchmark under similar conditions, on the same (reasonably large) dataset. Data-driven deep learning models enjoy the feature of “data synergy”, where larger and more diverse data leads to stronger and more robust models (Fang et al., 2022; Kratzert et al., 2021; Pasquiou et al., 2022; Yang et al., 2023b). Thus small-data comparison results may not be valid for a case with more data. Thus far, on the CAMELS dataset (Addor et al., 2017; Newman et al., 2014), both Kratzert et al. (2019) and Feng et al. (2021) reported very similar metric Nash-Sutcliffe model efficiency coefficient (NSE) values (Nash and Sutcliffe, 1970) for LSTM — 0.72 for 571 basins with the NLDAS forcing alone, making this a reliable benchmark that has thus far not been exceeded by other models. Sun et al. (2021) reported comparable results using GraphWaveNet, although with different training periods and ensemble setups. Furthermore, Kratzert simultaneously employed multiple forcing dataset (NLDAS, Maurer, and Daymet) for LSTM and obtained a Kling-Gupta model efficiency coefficient (KGE) (Gupta et al., 2009) of 0.80, setting a record on this dataset unmatched by any other model.

In this study, we investigate the performance of the Transformer architecture in rainfall-runoff modeling compared to LSTM using the CAMELS dataset. We analyze the performance of single models and ensembles for both architectures, and examine the models’ abilities to handle single and multiple forcings. This approach aims to establish a reference point for future studies to compare, enhancing our understanding of these models in complex scenarios. Our findings contribute to the understanding of the strengths and limitations of both LSTM and Transformer models in hydrological predictions, and highlight the potential of the Transformer as an alternative and scalable solution for hydrologic modeling.

2. Data and methods

2.1. Datasets

In this paper, we utilized the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) dataset (Addor et al., 2017; Newman et al., 2014), which includes basin-averaged daily data from 671 catchments across the conterminous United States (CONUS) characterized by minimal anthropogenic disturbances. The catchment attributes encompass an array of characteristics such as topography, soil properties, and geological aspects. Furthermore, CAMELS provides daily meteorological forcing inputs derived from three distinct gridded data products, namely Daymet (Thornton et al., 1997), Maurer (Maurer et al., 2002), and the North American Land Data Assimilation System (NLDAS) (Xia et al., 2012).

2.2. Vanilla (basic) Transformer models

The Transformer model, as first introduced in the paper “Attention is all you need” by Vaswani et al. (2017), is a neural network architecture for sequential data processing. The Transformer model consists of an encoder and a decoder. The encoder has a number (n_{layer}) of stacked encoding layers (“stacked” means the output of one layer becomes the input to the next one), each of which consists of a self-attention layer and a position-wise fully connected layer, while the decoder has only a simple position-wise linear layer. The critical mechanism within the encoder is self-attention, which computes the weighted sum of all input features. The equations for one of the stacked encoding layers are shown

below and explained one by one.

$$Q = x * D(W_q) \quad (1)$$

$$K = x * D(W_k) \quad (2)$$

$$V = x * D(W_v) \quad (3)$$

$$a = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (4)$$

$$c = a * V \quad (5)$$

$$u' = c * W_o \quad (6)$$

$$u = \text{BatchNorm}(x + D(u')) \quad (7)$$

$$z' = W_2 * D(\text{GELU}(W_1 * u)) \quad (8)$$

$$z = \text{BatchNorm}(u + z') \quad (9)$$

The inputs x to the attention layer have two dimensions — the sequence length (n) and a hidden-size dimension (d_k). In Eqs (1)–(3), the layer computes three sets of linear transformations, called Query, Key, and Value vectors (Q , K , and V), with respective learnable weights W_q , W_k , and W_v , all with the dimensions (d_k , d_k). These position-wise transformations (or matrix multiplications) mix information along the hidden-size dimension, not along the sequence length dimension, while applying the dropout operator $D()$. To mitigate overfitting, a dropout mask with a ratio of 0.5 is applied to W_q , W_k , and W_v . Eq (4) computes the dot product of Query and Key, and obtains a matrix of the size (n , n) which tabulates the similarity between each Query-Key pair. It then scales the calculations by $\sqrt{d_k}$, before applying the softmax operation along the sequence dimension. The output a is the above-mentioned attention weight while c is the attention-weighted values, called “contexts”. The model is called “multiheaded” because multiple sets of Q , K , and V are computed and their results c are applied through a linear layer as described in Eq. (6). Eqs. (7)–(9) apply additional linear layers with activation functions and residual connections to enhance training. z' is a feed-forward neural network (FFN) consisting of two linear transformations with a Gaussian Error Linear Unit (GELU) activation function in between. z includes a residual connection and batch normalization, where the elements along the batch dimension are normalized. As described earlier, Eqs. (1)–(9) are repeated n_{layer} times and the outputs of one layer serves as the inputs to the subsequent layer. The dimensional descriptions here all ignore the batch dimension (a collection of instances to compute a loss value and update the weights) which is in practice computed in parallel. The sequence length (n), the number of heads (h) and the hidden size (d_k) are hyperparameters to be tuned using the validation dataset.

Eqs. (4)–(5) can be interpreted as weighing every token in the sequence to make a combined prediction at a given location. We observe that, unlike RNNs which would naturally put more weight towards adjacent tokens, the sense of adjacency is lost for the attention layer — for prediction location i , all input tokens are treated equally, regardless of whether they are close to or far from i . The larger focus to adjacent tokens, if it exists in the training dataset, is completely obtained from data. Furthermore, any relational structure in the sequence dimension is not modeled — the softmax operator in Eq. (4) is the only operator that mixes information over the sequence length, as all the other operators are calculated in parallel for each token in the sequence. This setup is reasonable in language modeling where inversion structures are common, but may not be optimal if the proximity is important (as in natural physical processes). However, stacking many layers of attention sequentially as done in the Transformer could enable the modeling of some sequential structures.

The initial input to the model, X , is of dimension (n , n_X), which is

transformed by an embedding function. It includes three parts: a linear layer transformation of the inputs, a “positional embedding” (Eqs. (10)–(11)), and a “temporal embedding” (Eqs. (12)–(13)). These three components are directly summed to obtain the input x in Eqs. (1)–(3) which is then fed into the attention layers described above. The embeddings are added because the Transformer does not inherently account for the positional information. The positional encoding uses sine and cosine functions to create a unique encoding for each position, allowing the model’s self-attention mechanism to maintain the sequence order in context (Vaswani et al., 2017):

$$P_{k,2i} = \sin\left(\frac{k}{10000^{2i/d}}\right) \quad (10)$$

$$P_{k,2i+1} = \cos\left(\frac{k}{10000^{2i/d}}\right) \quad (11)$$

where k is the position in the sequence, d is the dimension, and i is the number of columns in the embedding matrix.

Furthermore, in the time series data, positional embedding alone can hardly reflect the seasonality information. Hence, hierarchical global timestamp information (weekly, monthly, yearly) is used to encode seasonality and long-term ordinal information (Zhou et al., 2021). This temporal embedding calculates and normalizes the day of the week, day of the month, and day of the year for each time period to a range of -0.5 to 0.5 :

$$d_i(k) = t_i(k)/N_i - 0.5 \quad (12)$$

$$T_e(k) = \oplus_{i \in \text{time_features}} d_i(k) \quad (13)$$

where t_i is the value of time feature i at position k in the sequence; for example, day of the week, day of the month, or day of the year. N_i is the total number of values for the time feature i ; for example, for the day of the year i , N_i would be 365. d_i is the normalized value for each time feature. $T_e(k)$ indicates the temporal embedding at position k . The ‘ $\oplus$ ’ symbol denotes concatenation, meaning it concatenates the time features into a single vector at the last dimension.

Positional and temporal embeddings are added to the input embeddings to form the input to the Transformer layer.

$$x = E_X(X) + E_P(P) + E_T(T) \quad (14)$$

where E_X , E_P , and E_T are the learned linear layer transformations of the input embedding, positional embedding, and temporal embedding layers, respectively, projecting the inputs to the model hidden dimension.

2.3. The Modified Transformer Model with Convolutional Embeddings

As a variant of the Transformer model, we added a one-dimensional convolutional embedding layer just before the attention layer to produce relational features in the time dimension. In this embedding layer, two stacked convolution sub-layers were introduced, with residual connections between them, and their outputs were fed into a linear layer. In each convolutional sub-layer, the time sequence length dimension gets convolved and, as such convolutions are non-recurrent, the model does not need to go through time steps in order to represent the temporal relational structures. The convolutional sub-layers have a dilation of 1, a stride of 1, ReLU as the activation functions, and a backward-focusing kernel to ensure that inputs from the future do not get used to make a prediction of the current time step. The kernel width, hidden sizes, and number of convolutional layers were set as hyperparameters that were tuned along with the hyperparameters of the attention layers. The outputs of the whole convolutional embedding layer were, along with the positional and temporal embeddings, added to the input embeddings just as in Eq. (14).

2.4. LSTM Models and SAC-SMA Models

In order to impartially evaluate the Transformer model's performance, we compared its results with those of LSTM and the Sacramento Soil Moisture Accounting (SAC-SMA) conceptual model (Anderson and McDonnell, 2005; Burnash et al., 1973), and used the latter two as benchmarks. We downloaded the SAC-SMA dataset from HydroShare (Kratzert et al., 2019), and set the same test time for all models to ensure a balanced comparison. This approach helps provide a thorough and fair assessment of each model's performance capabilities. The LSTM model's configurations were based on Kratzert et al. (2021), with the model's hyperparameters set to 30 epochs, a sequence length of 365, a hidden size of 256, and a dropout rate of 0.4.

The LSTM model from Kratzert et al., 2019 was originally evaluated on 531 basins. To broaden our insights into the impacts of a single forcing dataset on the entire CAMELS dataset and ensure a fair comparison, we retrained their model on the full set of 671 basins with the single NLDAS forcing dataset. We further attempted to incorporate time stamp information as inputs into the LSTM model (data not shown). However, this did not lead to any improved performance, suggesting it does not introduce new information to LSTM (see Fig. 1).

2.5. Experiments and Model Evaluation

To enable a comprehensive comparison with various benchmarks, we utilized data from all 671 CAMELS basins. To be consistent with previous benchmark experiments, we employed both single and multi-forcing datasets, in the same manner as in the benchmark (Kratzert et al., 2021). Initially, we applied a single forcing dataset derived from NLDAS, referred to as the 'single-forcing' experiment. Subsequently, we conducted a multi-forcing analysis using forcing data from Daymet, Maurer, and NLDAS. For this analysis, our scope was narrowed to 531 basins. It should be noted that the variable selection and settings for the model input data were chosen to be consistently aligned with those employed by Kratzert et al. (2021).

For all models, the data used for the training period was from 1 October 1999 to 30 September 2008, while the data used for the testing period was from 1 October 1989 to 30 September 1999. During the training period, the weights were optimized using the Adam optimizer with a learning rate of 0.0001.

To accurately compare different model architectures and

hyperparameters, we used one specific seed in Fig. 2. Thus any differences in model performance can be fully attributed to the specific architectural or hyperparameter variations. To increase the robustness of the analysis, we employed an ensemble approach, using ten simulations with different random seeds for each of the model architectures. The ensemble-averaged discharge for each model architecture is what is presented in the results here, as it not only helps to capture the variation in results due to randomness, but also provides more stable performance estimates.

We evaluated model performance using several metrics, including the Nash-Sutcliffe model efficiency coefficient (NSE) (Nash and Sutcliffe, 1970) and the Kling-Gupta model efficiency coefficient (KGE) (Gupta et al., 2009). We also considered the percent bias of the top 2 % peak flow range (FHV) and the bottom 30 % low flow range (FLV), which respectively highlight model performance for peak flows and baseflow (Yilmaz et al., 2008).

3. Results and discussion

The SAC-SMA model had the lowest performance across all experiments, and aligned consistently with the results of Feng et al. (2020) and Kratzert et al. (2021). For the single-forcing CAMELS benchmark (671 basins), the vanilla Transformer was outperformed by LSTM (Table 1; Fig. 2). Overall, the vanilla Transformer fell behind LSTM in all metrics, although not by much. Looking at Kling-Gupta Efficiency, the vanilla Transformer achieved a value of 0.71, compared to 0.73 for the LSTM. These results suggest that, without modification, the vanilla Transformer is missing some critical ability to simulate hydrologic processes.

The vanilla Transformer's under-performance is a curious case as it has been widely recognized that "attention is all you need" (Vaswani et al., 2017) in sequential modeling, and we have several interpretations of the results presented here. First, it is possible that the dataset size is too small here and advantages for the Transformer could emerge for larger quantities of data. Second, the natural hydrologic process is a "Markovian" system (Grey Nearing, personal communication) where the states at the current time step, rather than more remotely-in-the-past steps, completely determine the system's trajectory for future time steps (along with the forcings). To be more concrete, the soil moisture today, rather than that from any previous days, would have far more of an impact on tomorrow's streamflow discharge. This is in strong contrast to human languages, where the flexibility of word order often does not change the context, which would favor the attention-based Transformer architecture. Third, the accumulation of water and its nonlinear interactions makes memory effects important, but the Transformer does not have memory and is not necessarily skilled at capturing the effects of memory. Regardless of the reason, the results mean that the vanilla Transformer is not optimal for streamflow predictions at the very least, and further changes are likely needed in order to use it for modeling natural systems.

On the other hand, the modified Transformer demonstrates performance metrics that are comparable to or slightly surpass those of the LSTM. However, it exhibits greater variability among ensemble members, indicated by the standard deviation of the KGE metric: ± 0.003 for the LSTM and ± 0.007 for the modified Transformer. Its KGE (0.74) is slightly higher than LSTM (0.73), and the differences in FLV and FHV from LSTM's values are too small to call an advantage, considering their variability. As to be discussed below, while these differences are small, we simply should not expect larger differences as the possible room of improvement may be very small at this stage. The ensemble standard deviation of KGE is 0.003 with LSTM and 0.007 with the modified Transformer. The LSTM has a smaller ensemble standard deviation for FHV than the modified Transformer, while the opposite is true for FLV. The ensemble standard deviation of median FHV is 0.58 for LSTM and 2.94 for the modified Transformer, while this value for the FLV is 8.15 for the LSTM and 4.24 for the modified Transformer. This suggests that while we obtain very similar overall metrics, the LSTM and the modified

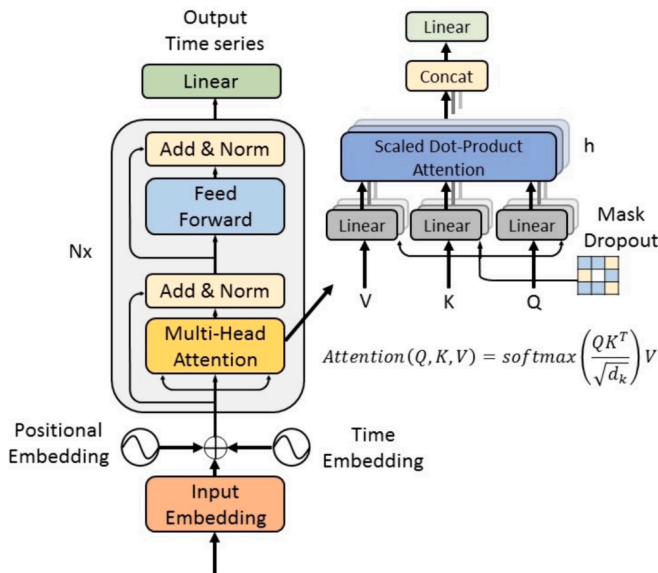


Fig. 1. The base Transformer model structure (adapted from Fig. 1 in Vaswani et al., 2017) used in this paper.

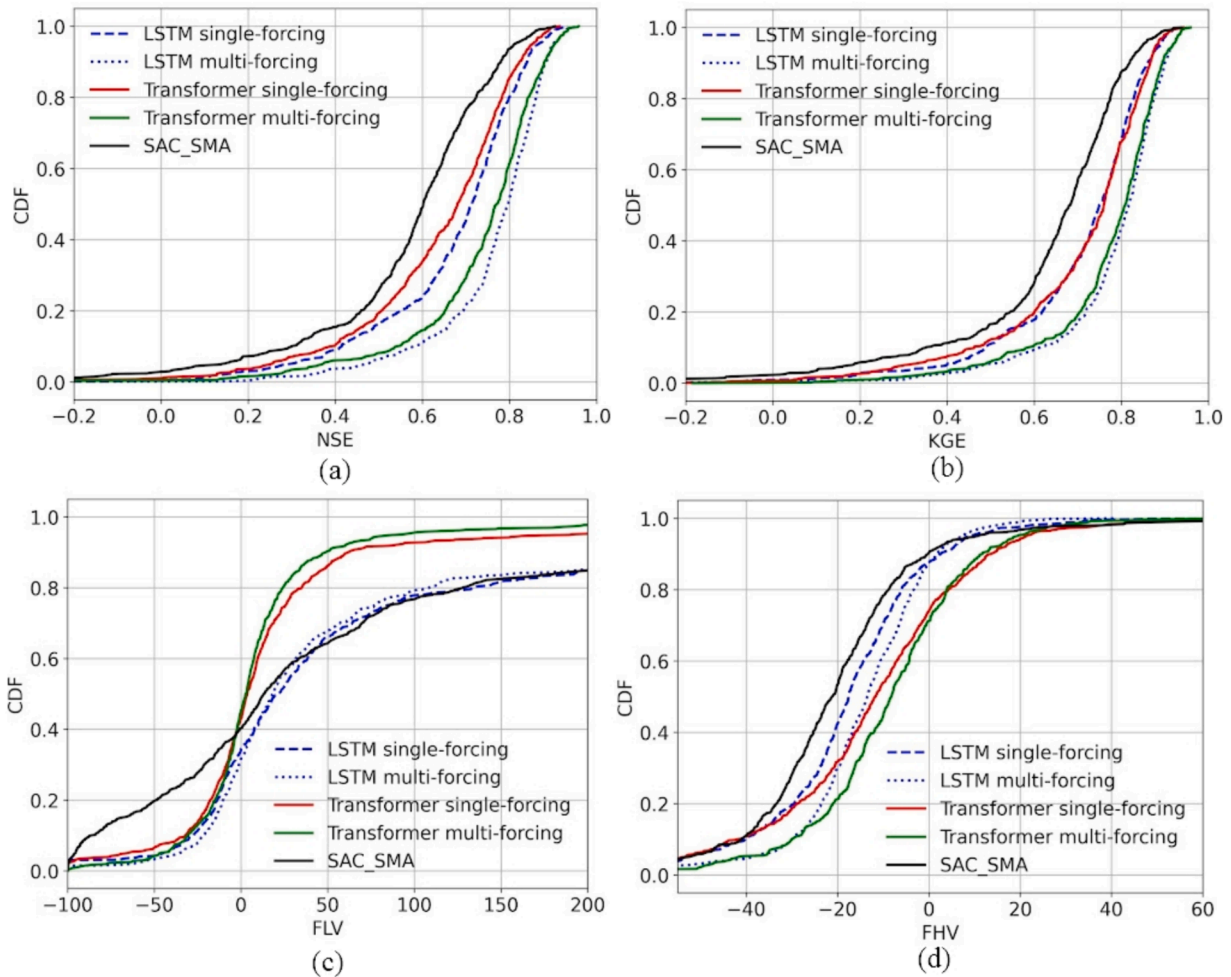


Fig. 2. Comparative analysis of Cumulative Density Function (CDF) across various models including Long Short-Term Memory (LSTM) and modified Transformer deep learning models, and the conceptual Sacramento Soil Moisture Accounting (SAC-SMA), with units in mm/day and one specific seed (rather than a random seed). The model encompasses single and multi-forcing data for models. The figure depicts the following comparisons: (a) Nash-Sutcliffe Efficiency (NSE) vs CDF, (b) Kling-Gupta Efficiency (KGE) vs CDF, (c) Low flow percent bias (FLV) vs CDF, and (d) High flow percent bias (FHV) vs CDF. Single-forcing models were implemented on a set of 671 basins in the CAMELS dataset, whereas multi-forcing models were applied to a subset of 531 basins from that dataset.

Table 1

Comparative performance metrics for single and multi-forcing experiments with LSTM, vanilla Transformer, and modified Transformer models. We conducted an evaluation of a single forcing on 671 basins and multi-forcing on 531 basins, employing the LSTM model results from [Kratzert et al., 2019](#), originally evaluated on 531 basins. To broaden our insights into the impacts of single-forcing on the entire CAMELS dataset and make a fair comparison, we retrained their model on an expanded set of 671 basins with single NLDAS dataset. These numbers are only very slightly different from [Kratzert et al., 2019](#). The means for Kling-Gupta Efficiency (KGE), high flow percent bias (FHV), and low flow percent bias (FLV) are averages from the 10 different ensemble members, each with a different random seed, while the standard deviations (std) for KGE, FHV, and FLV are calculated for the ensemble members.

	Forcing: NLDAS			Forcing: Multi-forcing		
	LSTM	Vanilla Transformer	Modified Transformer	LSTM	Vanilla Transformer	Modified Transformer
KGE (mean $\pm$ std)	0.73 $\pm$ 0.003	0.71 $\pm$ 0.007	0.74 $\pm$ 0.007	0.80 $\pm$ 0.004	0.77 $\pm$ 0.016	0.80 $\pm$ 0.007
FHV (mean $\pm$ std)	-17.49 $\pm$ 0.58	-26.66 $\pm$ 2.83	-18.00 $\pm$ 2.94	-11.91 $\pm$ 0.549	-21.54 $\pm$ 2.64	-9.19 $\pm$ 4.01
FLV (mean $\pm$ std)	-2.82 $\pm$ 8.15	3.31 $\pm$ 2.34	2.28 $\pm$ 4.24	2.57 $\pm$ 4.072	0.77 $\pm$ 1.65	2.72 $\pm$ 2.41

Transformer preferentially address different parts of the hydrograph in this experiment. LSTM more reliably focuses on the high-flow regime (quantified by the smaller ensemble standard deviation of FHV) than the modified Transformer, but the latter can better capture the long-term dependence (quantified by ensemble standard deviation of FLV representing groundwater baseflow). It seems there is some tradeoff for the different flow regimes.

The multi-forcing experiment generally shows similar patterns: the vanilla Transformer falls behind the other two models, which have very

similar ensemble-mean performance metrics but different ensemble standard deviations. The high KGE (0.80) and FHV (9.19, which is slightly better than that of the LSTM) demonstrate that the modified Transformer, like the LSTM, can effectively integrate different forcing datasets — a capability not shown by any other model architecture. Just as in the single-forcing NLDAS experiment, the modified Transformer has a larger stochastic variability (quantified by ensemble standard deviation) for FHV but smaller variability for FLV. Because both FHV and FLV have improved compared to the single-forcing experiment, the

modified Transformer was able to utilize the short-term and long-term dependencies of multiple forcing datasets. For one particular ensemble member (based on different random seeds), the cumulative density plot shows very similar curves between the modified Transformer and LSTM models.

The high agreement between the two model architectures, both of which are state of the art, suggests that we are likely at or very close to the predictive limit of the CAMELS dataset for this test (temporal test, training in one time period and testing in another). We suspect that unless we bring in new information, it is highly unlikely for any other models to produce noticeable advantages beyond these two models on this dataset, for the tests presented here. Errors with forcings, basin shapes, attributes, and discharge data are likely the remaining factors preventing higher performance. It should be mentioned that for another test, e.g., prediction in ungauged regions or spatial extrapolation, physics-informed hybrid models (called differentiable models (Shen et al., 2023)) can actually outperform LSTM (Aboelyazeed et al., 2023; Feng et al., 2022; Feng et al., 2023a; Tsai et al., 2021). Moreover, several issues surrounding the CAMELS dataset include using basin-average attributes that cannot resolve subbasin-level spatial heterogeneity, using daily precipitation that does not represent hourly rainfall intensity, the presence of major reservoirs in a fraction of basins, and the inclusion of some overly large basins.

Nevertheless, exactly because the Transformer algorithm does not have time integration, it can be trained in a highly parallel fashion and is suited to learning from large amounts of data. As the amount of data and the amount of neurons increase, it is possible to observe emergent behaviors of intelligence (Bubeck et al., 2023). This is a property that is worth further exploring in future studies in hydrology and geosciences. We leave to future work the question of how to incorporate more data with the modified Transformer, and the testing of this architecture on spatial extrapolation (for data-scarce scenarios) (Feng et al., 2021) and temporal extrapolation (for multidecadal projection of trends).

While some studies claim that Transformer models surpass LSTM models in performance, their evaluations are often limited to small datasets (Abed et al., 2022; Amanambu et al., 2022; Ghobadi and Kang, 2022), forecasting experiments (Yin et al., 2022), or still incorporate a mix of recurrent neural networks and attention mechanisms (Koya and Roy, 2023) (hereafter called KR23). KR23 evaluated the global Caravan dataset using the TFT model, yet several aspects of their approach hinder direct comparison. Firstly, KR23's comparison on a global dataset is valuable and welcomed, as they reported median KGE of 0.705 on 2610 basins for LSTM (basin-by-basin training) across the entire Caravan dataset, but did not provide results for the CAMELS dataset. When LSTM was benchmarked on a similar global dataset in another paper (Feng et al., 2023b) but was trained and tested for all basins simultaneously, it obtained a median KGE of 0.78 for 1675 basins with long-term records, far higher than the KGE of 0.679 reported in KR23, presumably due to using the recommended practice of training on all basins simultaneously (Fang et al., 2022; Kratzert et al., 2024). Secondly, with the addition of LSTM units in KR23's TFT, it is unclear how much the attention mechanisms helped performance in contrast to the LSTM units. Moreover, their model, being a hybrid of RNN and attention, still faces challenges with parallelization issues. Finally, they employed cubic spline interpolation on the streamflow data, which smooths the target data and undermines the comparability of model results. To validate our perspective, we employed the TFT model from PyTorch Forecasting (Beitner, 2020) on the CAMELS dataset and observed that the model's training speed was significantly slower, with each epoch taking 30 times longer than LSTM. We welcome the community to benchmark on shared datasets with transparent basin and input lists.

4. Conclusions

In this work, we compared a vanilla Transformer encoder and a modified Transformer to the current state-of-the-art model, LSTM, on

the CAMELS benchmark dataset. The vanilla Transformer seems to miss some critical functionality so that it is not optimal for simulating streamflow. The modified Transformer with no recurrent connection obtained slightly more favorable results (albeit only with a scale advantage) than LSTM. These results already represent rare competitive results to the LSTM in rigorous community-shared benchmarks. This means we can technically continue to search for better architecture to further improve its performance and suitability for natural physical systems, as the current setup may not yet be optimal. Nevertheless, the differences are overall small between the models, and we may already be very close to the optimum for this dataset with this test. On the one hand, we do not expect any architectural change to result in any significant improvement (to be more precise, on the order of 0.02 for KGE). An expansion of the dataset, e.g., better descriptors or forcing datasets will be required to obtain substantial prediction improvements. On the other hand, the modified Transformer architecture is a viable alternative to LSTM and may find advantages for larger datasets in the future. The Transformer architecture's advantages may not reside with sequential information extraction but with serving as a foundational model to capture the joint distribution, accumulate knowledge, and extract deep, abstract, and complex concepts. These advantages should be explored in future hydrologic and geoscientific research.

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CRedit authorship contribution statement

Jiangtao Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Yuchen Bian:** Methodology. **Kathryn Lawson:** Writing – review & editing. **Chaopeng Shen:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: CS and KL have financial interests in HydroSapient, Inc., a company which could potentially benefit from the results of this research. This interest has been reviewed by the University in accordance with its Individual Conflict of Interest policy, for the purpose of maintaining the objectivity and the integrity of research at The Pennsylvania State University.

Data availability

All data used for the analysis in this work is publicly available and is cited respectively.

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